

## PEQUOT LAKES, MN, USA

WMO: 727500

Lat: 46.600N

Lon: 94.317W

Elev: 1280

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 82-95

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 1             | -23.4      | -17.2 | -35.4                         | 0.8 | -22.8 | -29.2 | 1.1 | -16.4 | 12.3                  | 7.2  | 10.9 | 8.6  | 2.8                   | 300  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.6                   | 88.9            | 69.8 | 85.4 | 68.3 | 81.8 | 66.2 | 72.6                | 85.1 | 70.3 | 81.8 | 68.2 | 78.6 | 4.1                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.1                            | 108.3 | 80.6 | 66.1 | 101.0 | 77.1 | 64.2 | 94.4 | 74.6 | 37.1          | 85.3 | 35.0 | 82.0 | 33.2 | 78.6 | 79.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|-----|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
| 1%                |     | 2.5% | 5%                         | Mean  |      | Standard Deviation | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| Min               | Max | Min  | Max                        | Min   | Max  | Min                | Max                                                | Min   | Max        | Min   | Max        | Min   | Max        | Min   | Max   | DB  |
| (a)               | (b) | (c)  | (d)                        | (e)   | (f)  | (g)                | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)   |     |
| (4) 10.8          | 9.6 | 7.9  |                            | -30.5 | 94.4 | 5.2                | 4.9                                                | -34.2 | 97.9       | -37.3 | 100.8      | -40.2 | 103.6      | -43.9 | 107.1 |     |
| (5)               |     |      | DB                         | -32.0 | 75.4 | 5.3                | 2.7                                                | -35.8 | 77.3       | -39.0 | 78.8       | -42.0 | 80.3       | -45.8 | 82.2  | (p) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 40.1   | 9.1   | 14.1  | 26.7  | 40.7  | 54.7  | 63.5  | 68.3  | 66.5  | 55.3  | 42.6  | 26.3  | 12.0  |
|                                                                                  | DBStd       | 23.33  | 13.54 | 14.17 | 11.22 | 9.47  | 8.52  | 7.22  | 6.63  | 6.65  | 8.41  | 8.68  | 9.66  | 14.00 |
|                                                                                  | HDD50       | 5542   | 1267  | 1005  | 723   | 309   | 46    | 1     | 0     | 0     | 40    | 262   | 711   | 1178  |
|                                                                                  | HDD65       | 9433   | 1732  | 1425  | 1187  | 731   | 336   | 114   | 39    | 64    | 306   | 696   | 1160  | 1643  |
|                                                                                  | CDD50       | 1936   | 0     | 0     | 0     | 29    | 191   | 405   | 568   | 511   | 199   | 32    | 1     | 0     |
|                                                                                  | CDD65       | 352    | 0     | 0     | 0     | 1     | 15    | 68    | 142   | 110   | 16    | 0     | 0     | 0     |
|                                                                                  | CDH74       | 3568   | 0     | 0     | 0     | 16    | 232   | 721   | 1462  | 980   | 147   | 10    | 0     | 0     |
|                                                                                  | CDH80       | 1126   | 0     | 0     | 0     | 1     | 43    | 220   | 526   | 301   | 34    | 1     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.4    | 3.4   | 3.1   | 3.8   | 4.0   | 3.7   | 3.3   | 2.8   | 2.7   | 3.3   | 3.5   | 3.7   | 3.1   |
| Precipitation                                                                    | PrecAvg     | 28.0   | 0.7   | 0.6   | 1.3   | 2.1   | 3.2   | 4.3   | 4.1   | 3.1   | 3.1   | 3.1   | 1.3   | 0.9   |
|                                                                                  | PrecMax     | 33.7   | 1.9   | 1.9   | 3.5   | 5.2   | 7.6   | 7.5   | 7.6   | 6.3   | 7.1   | 6.0   | 4.1   | 2.1   |
|                                                                                  | PrecMin     | 20.8   | 0.1   | 0.0   | 0.4   | 0.8   | 1.2   | 1.5   | 1.9   | 0.9   | 0.7   | 0.4   | 0.2   | 0.3   |
|                                                                                  | PrecStd     | 3.3    | 0.5   | 0.4   | 0.9   | 1.0   | 1.5   | 1.6   | 1.6   | 1.5   | 1.3   | 1.6   | 0.9   | 0.5   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 39.9  | 44.2  | 57.2  | 76.1  | 84.8  | 90.9  | 97.1  | 91.3  | 84.3  | 74.7  | 54.3  | 43.0  |
|                                                                                  |             | MCWB   | 34.5  | 36.8  | 45.1  | 57.3  | 65.8  | 69.3  | 73.7  | 72.4  | 66.6  | 59.4  | 49.1  | 36.8  |
|                                                                                  | 2%          | DB     | 35.0  | 39.9  | 49.2  | 69.5  | 80.0  | 86.1  | 90.1  | 87.1  | 77.8  | 67.1  | 47.8  | 36.7  |
|                                                                                  |             | MCWB   | 30.9  | 34.0  | 40.2  | 53.5  | 61.1  | 68.1  | 70.5  | 70.7  | 64.7  | 54.9  | 42.8  | 32.3  |
|                                                                                  | 5%          | DB     | 31.5  | 36.1  | 44.8  | 63.2  | 76.0  | 81.6  | 85.9  | 83.2  | 73.6  | 62.0  | 43.2  | 33.0  |
|                                                                                  |             | MCWB   | 28.8  | 31.8  | 37.7  | 49.1  | 59.2  | 65.4  | 68.6  | 67.8  | 62.6  | 52.5  | 39.0  | 30.2  |
|                                                                                  | 10%         | DB     | 28.9  | 33.1  | 40.5  | 57.5  | 71.7  | 77.7  | 82.0  | 79.6  | 69.5  | 57.1  | 39.4  | 30.7  |
|                                                                                  |             | MCWB   | 26.6  | 29.9  | 35.0  | 45.4  | 57.2  | 63.4  | 66.9  | 65.8  | 60.4  | 48.3  | 35.6  | 28.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 34.7  | 37.2  | 46.7  | 60.8  | 67.3  | 73.1  | 75.7  | 75.0  | 70.2  | 60.4  | 50.8  | 38.6  |
|                                                                                  |             | MCDB   | 39.5  | 43.2  | 55.2  | 70.8  | 82.7  | 85.8  | 92.6  | 87.2  | 78.7  | 69.9  | 53.3  | 42.8  |
|                                                                                  | 2%          | WB     | 31.1  | 34.3  | 41.8  | 55.2  | 64.0  | 69.6  | 72.9  | 72.4  | 67.2  | 57.7  | 43.5  | 32.8  |
|                                                                                  |             | MCDB   | 34.7  | 39.0  | 48.7  | 67.4  | 76.0  | 81.7  | 86.3  | 84.1  | 74.3  | 63.3  | 47.8  | 36.0  |
|                                                                                  | 5%          | WB     | 29.2  | 32.1  | 38.1  | 50.7  | 61.4  | 67.2  | 70.5  | 70.0  | 64.5  | 53.8  | 39.3  | 30.6  |
|                                                                                  |             | MCDB   | 31.6  | 36.0  | 44.1  | 62.0  | 72.8  | 79.2  | 82.9  | 80.3  | 71.8  | 62.2  | 43.0  | 32.9  |
|                                                                                  | 10%         | WB     | 26.8  | 30.0  | 35.3  | 46.2  | 58.7  | 64.9  | 68.1  | 67.9  | 61.8  | 49.1  | 35.9  | 28.7  |
|                                                                                  |             | MCDB   | 29.1  | 33.3  | 40.2  | 56.7  | 69.1  | 75.5  | 79.4  | 76.9  | 68.2  | 56.7  | 39.3  | 31.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 19.0  | 18.8  | 17.8  | 21.7  | 23.6  | 22.0  | 21.6  | 21.0  | 19.9  | 18.9  | 13.4  | 16.8  |
|                                                                                  |             | MCDBR  | 17.3  | 19.2  | 22.2  | 31.6  | 28.6  | 27.1  | 26.2  | 25.5  | 24.6  | 27.0  | 18.4  | 16.2  |
|                                                                                  |             | MCWBR  | 14.7  | 15.1  | 15.2  | 18.7  | 15.0  | 14.0  | 12.6  | 12.6  | 14.7  | 16.6  | 14.1  | 13.9  |
|                                                                                  | 5% WB       | MCDBR  | 16.8  | 18.2  | 20.8  | 27.6  | 24.3  | 23.8  | 24.0  | 22.2  | 21.2  | 25.3  | 17.3  | 15.8  |
|                                                                                  |             | MCWBR  | 14.4  | 14.5  | 14.6  | 17.4  | 14.1  | 13.4  | 13.2  | 12.1  | 14.4  | 16.7  | 14.1  | 13.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.256  | 0.286 | 0.308 | 0.357 | 0.377 | 0.376 | 0.379 | 0.386 | 0.355 | 0.311 | 0.283 | 0.263 |       |
|                                                                                  | taud        | 2.378  | 2.295 | 2.386 | 2.343 | 2.344 | 2.377 | 2.387 | 2.383 | 2.448 | 2.562 | 2.519 | 2.405 |       |
|                                                                                  | Ebn at Noon | 271    | 281   | 292   | 285   | 282   | 282   | 279   | 273   | 272   | 272   | 260   | 255   |       |
|                                                                                  | Edh at Noon | 22     | 30    | 32    | 37    | 38    | 37    | 36    | 35    | 30    | 23    | 20    | 19    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 506    | 862   | 1214  | 1526  | 1700  | 1860  | 1977  | 1662  | 1243  | 765   | 472   | 351   |       |
|                                                                                  | RadStd      | 57     | 87    | 120   | 131   | 123   | 125   | 84    | 140   | 93    | 115   | 48    | 43    |       |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   | (n)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (49 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page